

$$\dot{m} = \rho A v$$

Solid propellants

Homogeneous

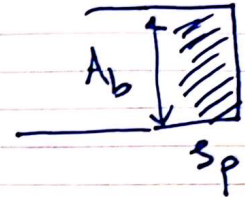
fuel and oxidizer
in the same molecule

Ex: double-base propellant
(nitroglycerin-nitrocellulose)

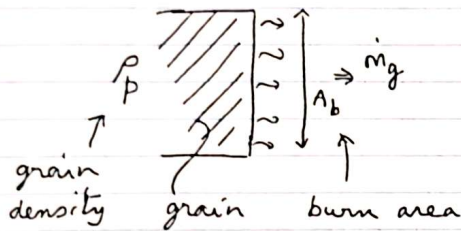
Heterogeneous

Fuel and oxidizer
grossly mixed

Ex: composite
(plastic-like binder: HTPB)
oxidizing crystals: AP, AN,
NP, KP, KN
Metal additives: Al, Mg, B



Solid propellants are observed to regress at a
uniform rate during their burn (regression rate r)



$$\dot{m}_g = \rho_p A_b r$$

r : burn rate

regression rate

$$\dot{m}_g = \rho_p A_b r$$

The regression rate is related to pressure as follows:

$$r = a p_o^n \quad (12.25)$$

n - pressure exponent (independent of temperature)

a - depends on initial temperature of the grain (T_p):

$$a = \frac{A}{T_i - T_p} \quad (12.26)$$

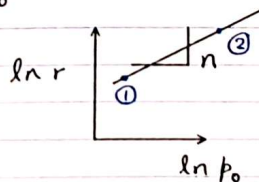
{ A, T_i are constants }

$$(12.25) \Rightarrow \ln r = \ln a + n \ln p_o$$

$$\ln r_2 = \ln a + n \ln p_{o2}$$

$$\ln r_1 = \ln a + n \ln p_{o1}$$

$$(\ln r_2 - \ln r_1) = n(\ln p_{o2} - \ln p_{o1})$$



$$r = a p_o^n$$

$$a = \frac{A}{T_i - T_p}$$

"Propellant temperature"

$$\dot{m}_g = \rho_p A_b \frac{A}{T_i - T_p} p_o^n$$

$$\Rightarrow n = \frac{\ln(r_2/r_1)}{\ln(p_{02}/p_{01})}$$

Quasi-steady model

$$\dot{m}_g = \dot{m}_t$$

$$\dot{m}_g = \rho_p A_b r = \rho_p A_b a p_0^n$$

$$\dot{m}_t = \frac{p_0}{\sqrt{RT_0}} \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} A_t \quad \{M_t = 1\}$$

Equating $\dot{m}_g$ to $\dot{m}_t$ yields the steady state operating pressure of the rocket motor

$$p_0 = \left[\frac{A_b}{A_t} \frac{a \rho_p}{\sqrt{\frac{\gamma}{RT_0} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}}} \right]^{\frac{1}{1-n}} \quad \text{See (12.29)}$$

$$\Rightarrow p_0 = \left(\frac{A_b}{A_t} a \rho_p c^* \right)^{\frac{1}{1-n}}$$

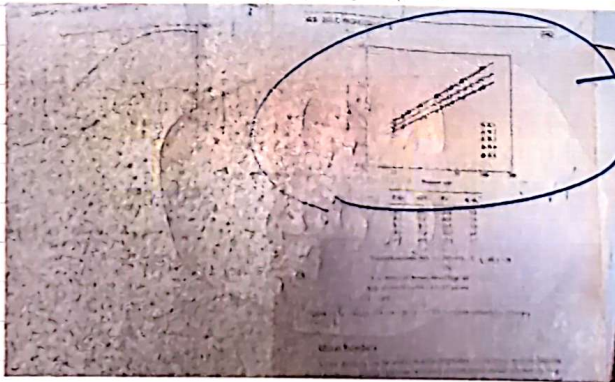
$$\text{Char. Velocity} \Rightarrow c^* = \frac{p_0 A_t}{\dot{m}}$$

Ex: Find $\left(\frac{A_b}{A_t} \right)$ required to reach a steady state

operating pressure of 14 MPa in a SRM with the following characteristics:

$$\rho_p = 1710 \text{ kg/m}^3, \gamma = 1.27, T_0 = 2220 \text{ K}, \bar{M} = 23,$$

R-4 propellant, see Fig 12.15 for burn data

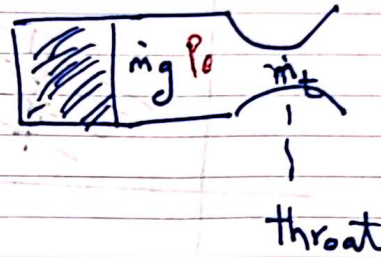
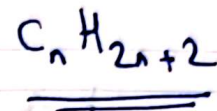
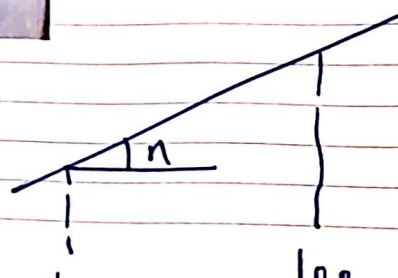


→ See this figure in announcement in Canvas

Fig. 12.15

$$\text{At } p_{01} = 10 \text{ atm}, r_1 = 0.07 \text{ cm/s}$$

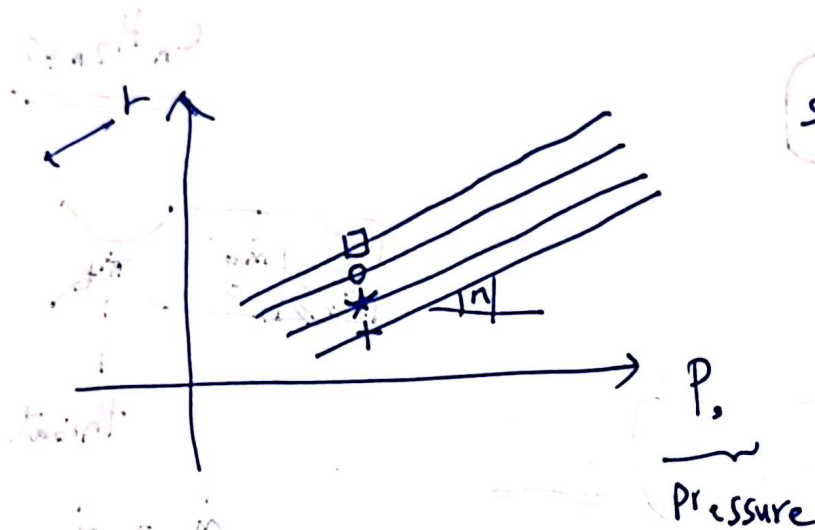
$$\text{At } p_{02} = 100 \text{ atm}, r_2 = 0.22 \text{ cm/s}$$



$$\dot{m}_g = \dot{m}_t$$

$$\rho_p A_b \frac{A}{T \cdot T_p} p_0^n =$$

burning
rate



symbols: Different
fuels

How to get n ?

$$r = a P^n \Rightarrow \ln r = \ln a + n \ln P$$

$$\ln r_1 = \ln a_1 + n \ln P_1$$

$$\ln r_2 = \ln a_2 + n \ln P_2$$

$$(\ln r_2 - \ln r_1) = n (\ln P_2 - \ln P_1)$$

$$\Rightarrow n = \frac{\ln r_2 - \ln r_1}{\ln P_2 - \ln P_1}$$

$$n = \frac{\ln\left(\frac{r_2}{r_1}\right)}{\ln\left(\frac{P_2}{P_1}\right)}$$

At $p_{02} = 100 \text{ atm}$, $r_2 = 0.22 \text{ cm/s}$

$$n = \frac{\ln\left(\frac{r_2}{r_1}\right)}{\ln\left(\frac{p_{02}}{p_{01}}\right)} = \frac{\ln\left(\frac{0.22}{0.07}\right)}{\ln\left(\frac{100}{10}\right)} = 0.4973 < 1$$

$$a = \frac{r_1}{p_{01}^n} = \frac{(0.07)(10^{-2})}{[10(101325)]^{0.4973}} = 7.2186(10^{-7}) \left\{ \frac{\text{m/s}}{\text{Pa}^{0.4973}} \right\} \text{ or } a = \frac{r_2}{p_{02}^n}$$

$p_0 = 14 \text{ MPa}$, $\rho_p = 1710 \text{ kg/m}^3$, $\gamma = 1.27$, $R = \frac{\bar{R}}{M} = 361.5 \frac{\text{J}}{\text{kg-K}}$

$$c^* = \sqrt{\frac{1}{\gamma} R T_0 \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma}{\gamma-1}}} = 1353.7 \text{ m/s}$$

$$p_0 = \left(\frac{A_b}{A_t} a \rho_p c^* \right)^{\frac{1}{1-n}}$$

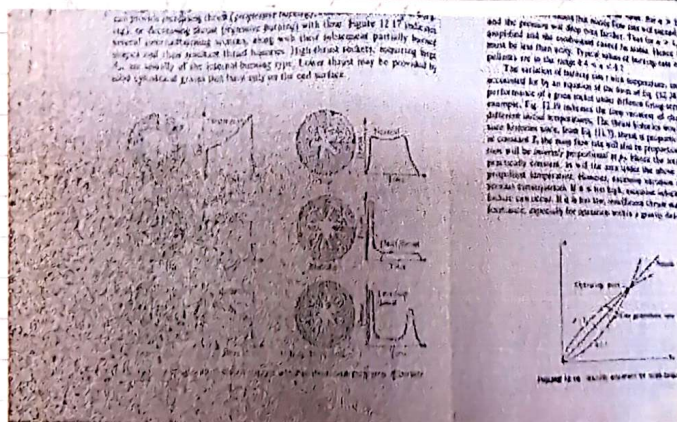
$$\Rightarrow \frac{A_b}{A_t} = \frac{p_0^{1-n}}{a \rho_p c^*} = 2341$$

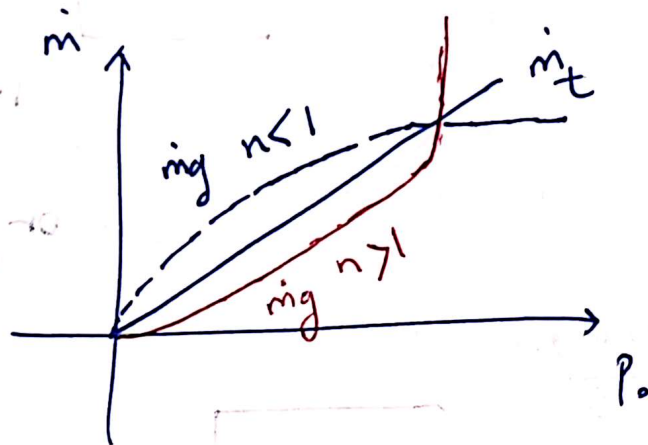
$$[r = a p_0^n = 7.2186(10^{-7}) [14(10^6)]^{0.4973}]$$

$\uparrow \frac{\text{m/s}}{\text{Pa}^{0.4973}} \quad \text{Pa}$

$$r = 0.002584 \text{ m/s} = 0.258 \text{ cm/s}$$

See Fig 12.17 for schematics of several internal burning grains and associated thrust histories





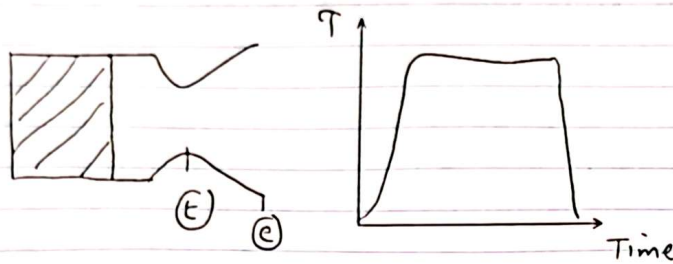
We know that $\dot{m}_g = \dot{m}_t$
 $\dot{m}_t$ is linear function of p_0

$$\dot{m}_g \propto p_0^n$$

$$p^n > p \quad n > 1$$

$$p^n < p \quad n < 1 \rightarrow \text{preferred}$$

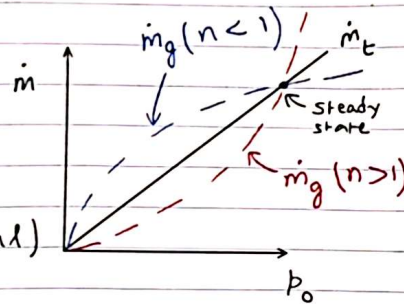
End burning grains burn neutrally.



Burning stability

$n > 1$: explosive behavior
 $n < 1$: stable behavior

$0.4 < n < 0.7$ (typical)



The sensitivity of burning rate to initial propellant temperature is defined as

$$\pi_r \stackrel{\text{def}}{=} \frac{1}{r} \left(\frac{\partial r}{\partial T_p} \right)_{P_0} = \left[\frac{\partial}{\partial T_p} (\ln r) \right]_{P_0} \quad (12.30)$$

$$\pi_r = \frac{1}{r} \left(\frac{\partial r}{\partial T_p} \right)_{P_0 = cte} = \frac{\partial \ln r}{\partial T_p} \Big|_{P_0 = cte}$$

$$r = a P_0^n, \quad \ln r = \ln a + n \ln P_0$$

$$\pi_r = \left[\frac{\partial (\ln r)}{\partial T_p} \right]_{P_0} = \frac{d(\ln a)}{dT_p} + 0$$

Since $a = \frac{A}{T_i - T_p}$, $\ln a = \ln A - \ln(T_i - T_p)$

$$\frac{d(\ln a)}{dT_p} = 0 + \frac{1}{T_i - T_p}, \quad \text{(or)} \quad \frac{1}{a} \frac{da}{dT_p} = \frac{1}{T_i - T_p}$$

$$r = a P_0^n \rightarrow \ln r = \ln a + n \ln P_0$$

$$\Rightarrow \frac{\partial \ln r}{\partial T_p} = \frac{\partial \ln a}{\partial T_p} \Big|_{P_0 = cte} + 0$$

$$\pi_r = \frac{1}{T_i - T_p} \quad (12.31)$$

$$0.002 \text{ K}^{-1} < \pi_r < 0.008 \text{ K}^{-1} \quad (\text{typical } \pi_r \text{ range})$$

The sensitivity of p_0 to initial grain temperature is defined as

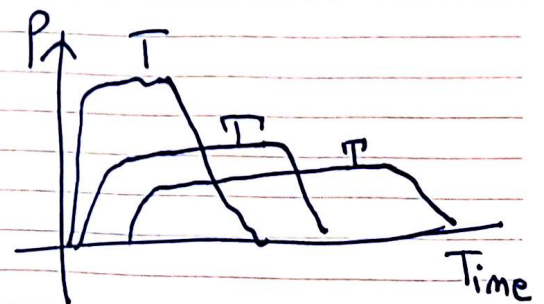
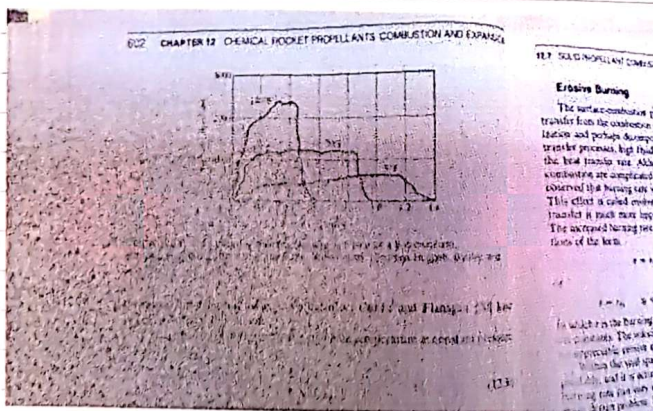
$$\pi_p = \frac{1}{p_0} \left(\frac{\partial p_0}{\partial T_p} \right)_{\frac{A_b}{A_t}, T_0} = \left[\frac{\partial}{\partial T_p} (\ln p_0) \right]_{\frac{A_b}{A_t}, T_0}$$

From $p_0 = \left(\frac{A_b}{A_t} a p_c c^* \right)^{\frac{1}{1-n}}$, and observing only a is initial temperature dependent, $a = \frac{A \rightarrow c^*}{T_0 - T_p}$

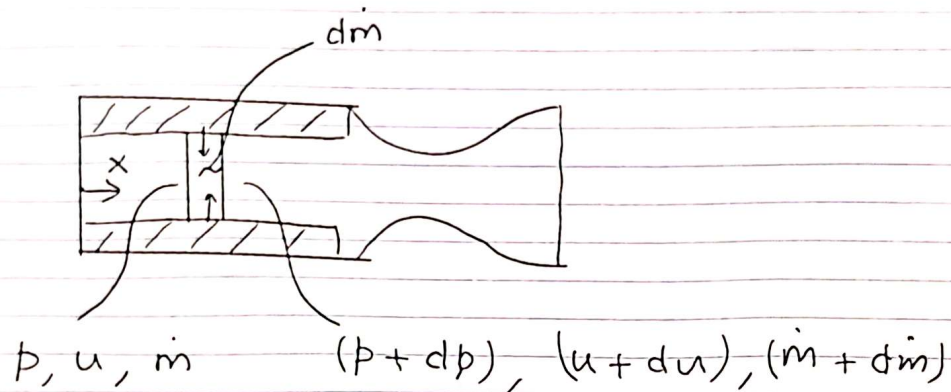
$$\left[\frac{\partial (\ln p_0)}{\partial T_p} \right]_{\frac{A_b}{A_t}, T_0} = \frac{1}{1-n} \frac{d(\ln a)}{dT_p} = \frac{1}{1-n} \pi_r \quad (12.32)$$

$$\pi_p = \frac{1}{1-n} \pi_r = \frac{1}{(1-n)(T_i - T_p)}$$

To lessen the sensitivity of p_0 to initial grain temperature, the propellant should have high T_i and low n . Fig 12.19 shows the impact of the initial grain temperature on p_0 .



Axial pressure variation



x - Momentum:

$$\cancel{\dot{m}u} + (d\dot{m})(0) + \cancel{pA} - (p + dp)A = (\dot{m} + d\dot{m})(u + du) \quad \text{neglect}$$

$$\uparrow \quad \quad \quad = \cancel{\dot{m}u} + u d\dot{m} + \dot{m} du + (d\dot{m})(du) \quad \text{2nd order}$$

$$\Rightarrow \boxed{-A dp = u d\dot{m} + \dot{m} du = d(\dot{m}u)} \quad \int$$

$$-\int A dp = \int d(\dot{m}u)$$

$$p = p_1 \quad \dot{m}u = 0$$

$$\Rightarrow A(p_1 - p) = (\dot{m}u)_x$$

$$\Rightarrow p = p_1 - \frac{(\dot{m}u)_x}{A} \quad (12.35) \quad \dot{m} = \rho u A \Rightarrow$$

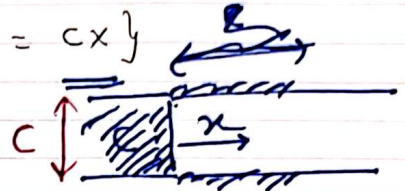
$$\text{Since } u = \frac{\dot{m}}{\rho A},$$

$$p = p_1 - \frac{1}{\rho} \left(\frac{\dot{m}}{A} \right)^2 \quad (12.36)$$

If c is the circumference of the grain, the mass flow

rate at any station x is

$$\dot{m} = \rho_p (cx) r \quad \{A_b = cx\}$$



$$p = p_1 - \frac{RT}{p} \left(\frac{\rho_p cx r}{A} \right)^2$$

$$\Rightarrow p = \frac{p_1}{2} \left[1 + \sqrt{1 - 4RT \left(\frac{\rho_p crx}{p_1 A} \right)^2} \right] \quad (12.37)$$

Pressure change in C.C